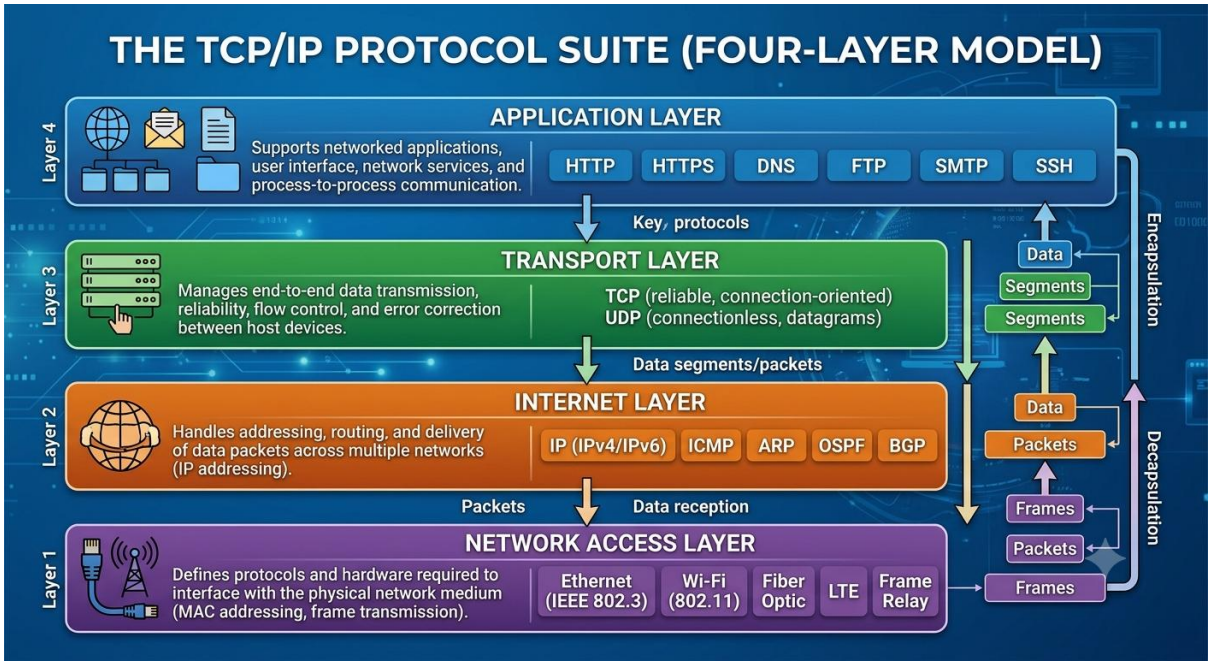
How the Internet Delivers a Web Page

1. TCP/IP Protocol Stack



2. Role of DNS

When the user enters www.amazon.in, the browser contacts a DNS server to convert the domain name into its corresponding IP address. This IP address helps the browser locate Amazon's web server.

3. HTTP Communication

The browser sends an **HTTP GET** request to the Amazon server. The server responds with the requested webpage, including HTML, CSS, JavaScript, images, and product information. Secure communication is provided using **HTTPS**.

4. TCP Reliability

TCP ensures reliable communication by:

- Establishing a connection (Three-Way Handshake)
- Sending data in sequence
- Detecting lost packets
- Retransmitting missing data

- Delivering data without errors

5. Architecture

Amazon follows a **Client–Server Architecture**.

- **Client:** User's web browser
- **Server:** Amazon's web servers

The browser requests information, and the server processes the request and sends the webpage back to the user.

6. HTTP vs FTP

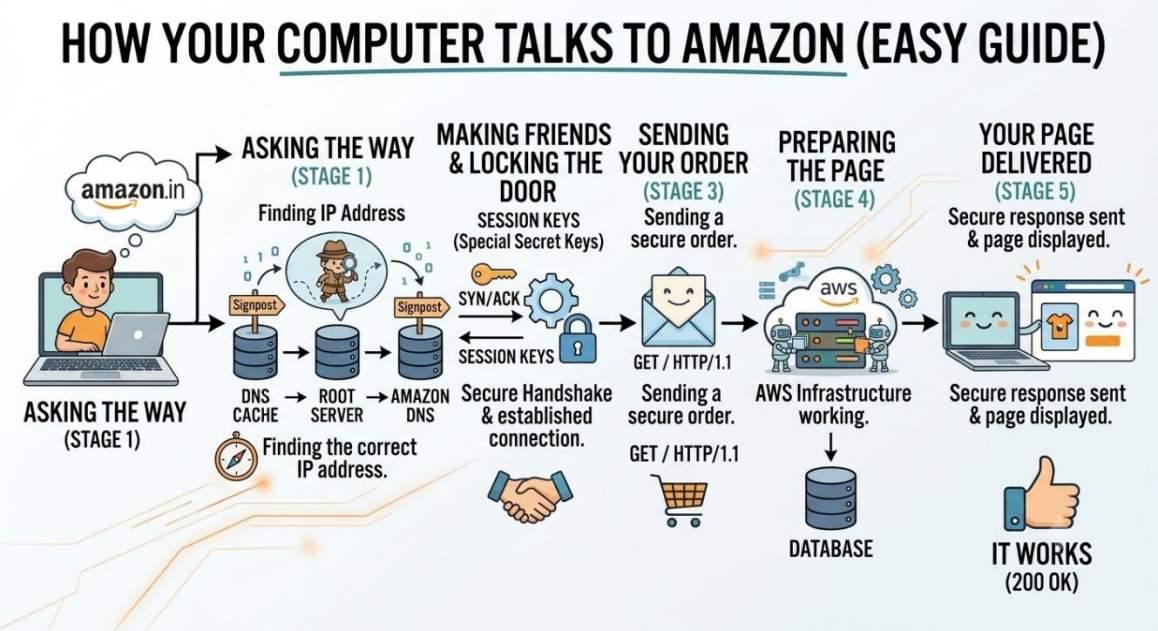
HTTP	FTP
Used to access web pages	Used to transfer files
Port 80	Ports 20 and 21
Stateless protocol	Stateful protocol
Used by web browsers	Used by FTP clients

7. Cookies

Cookies store user information such as login sessions and shopping cart items.

Example: When you add products to your cart on Amazon, cookies remember the selected items even if you browse different pages.

8. Flow Diagram



9.Conclusion

This assignment helped me understand how the Internet delivers a web page when a user enters a website address in a browser. It explained the roles of the TCP/IP protocol stack, DNS, HTTP, and TCP in establishing communication between the client and the web server. DNS converts the domain name into an IP address, while HTTP is used to request and receive web pages. TCP ensures that all data is delivered accurately, in the correct order, and without loss. The assignment also highlighted the importance of the client–server architecture, cookies, and the differences between HTTP and FTP. These technologies work together to provide a fast, reliable, and secure browsing experience. Understanding these application-layer protocols gives a clear picture of how websites function behind the scenes and why they are essential for everyday Internet activities such as web browsing, online shopping, email, and social media.